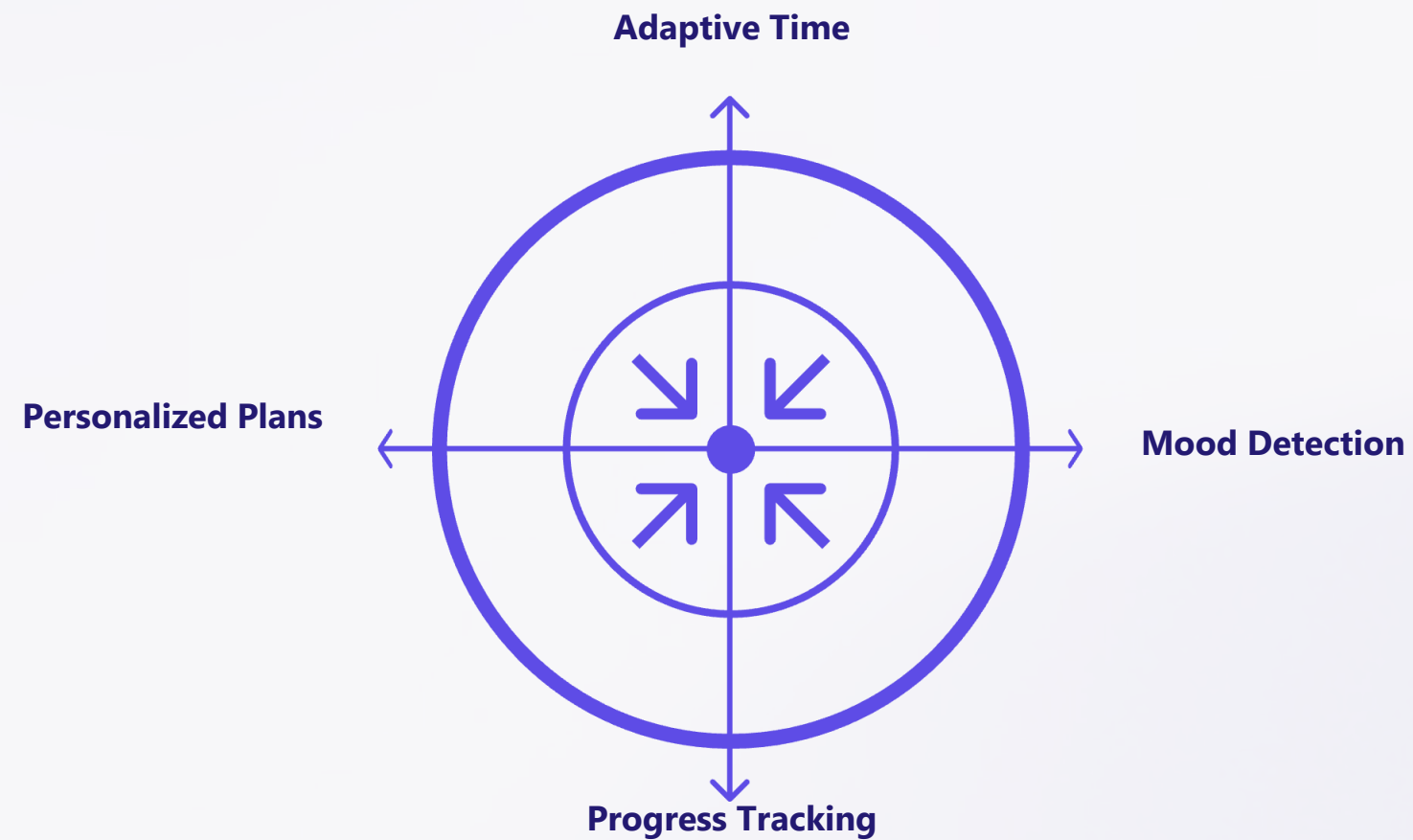


Nadav Hacham & Nadav Zimmerman

Practice Arena

A smart practice planner for guitar players that adapts to your time, mood, and progress



OUR STORY

Why We Built Practice Arena

As guitar players and teachers ourselves, we've experienced the challenge firsthand: creating effective practice plans that fit unpredictable schedules, fluctuating energy levels, and individual learning styles.

We built Practice Arena to solve this problem—not just for our students, but for ourselves. The app generates personalized practice sessions based on three key factors: how much time you have, how you're feeling, and what your practice history reveals about your progress.





 RESEARCH

Literature Review

Lozano-Murciego(2021)

Context Is Critical

Research shows that mood, available time, and motivation significantly impact practice effectiveness. Static lesson plans ignore these factors.

Zhang (2025)

Personalization Drives Engagement

Adaptive systems that tailor content to individual characteristics and recent performance improve engagement and reduce frustration.

Chen (2025) + Yang (2024)

Emotion Matters in Learning

Emotion-aware educational technologies positively influence motivation, cognitive performance, and persistence in skill acquisition.

Most existing practice apps focus on content delivery and performance scoring. Practice Arena fills the gap by dynamically adapting entire practice sessions to your unique context and behavioral patterns.

Competitor Review



Yousician

Focus: Real-time performance evaluation using audio analysis

Limitation: Fixed lesson paths driven by accuracy scores, not context or behavioral patterns



Melodics

Focus: Repetitive skill-building exercises with performance scoring

Limitation: Exercise-centric recommendations without dynamic session restructuring based on mood or time



Modacity

Focus: Manual practice organization and time tracking

Limitation: Users must structure their own sessions—no automated behavioral inference or adaptive planning

Practice Arena vs. The Competition

Core Capability	Practice Arena	Yousician	Melodics	Modacity
Adaptive Full Practice Plan	✓ Yes	✗ Lesson-based	✗ Exercise-based	✗ User-defined
Emotional & Context Awareness	✓ Mood + Time	✗	✗	✗
Behavioral Profiling	✓ Completion + Consistency	Limited	Limited	✗
Long-Term Practice Path	✓ Explicit model	✗	✗	✗

📄 Practice Arena adapts entire sessions rather than individual exercises, combining emotional context, time constraints, and behavioral data to build a dynamic, evolving practice strategy.

Why Practice Arena Stands Apart

Context-Aware Adaptation

We don't just recommend exercises; we adapt entire practice sessions based on your mood, time, and progress.



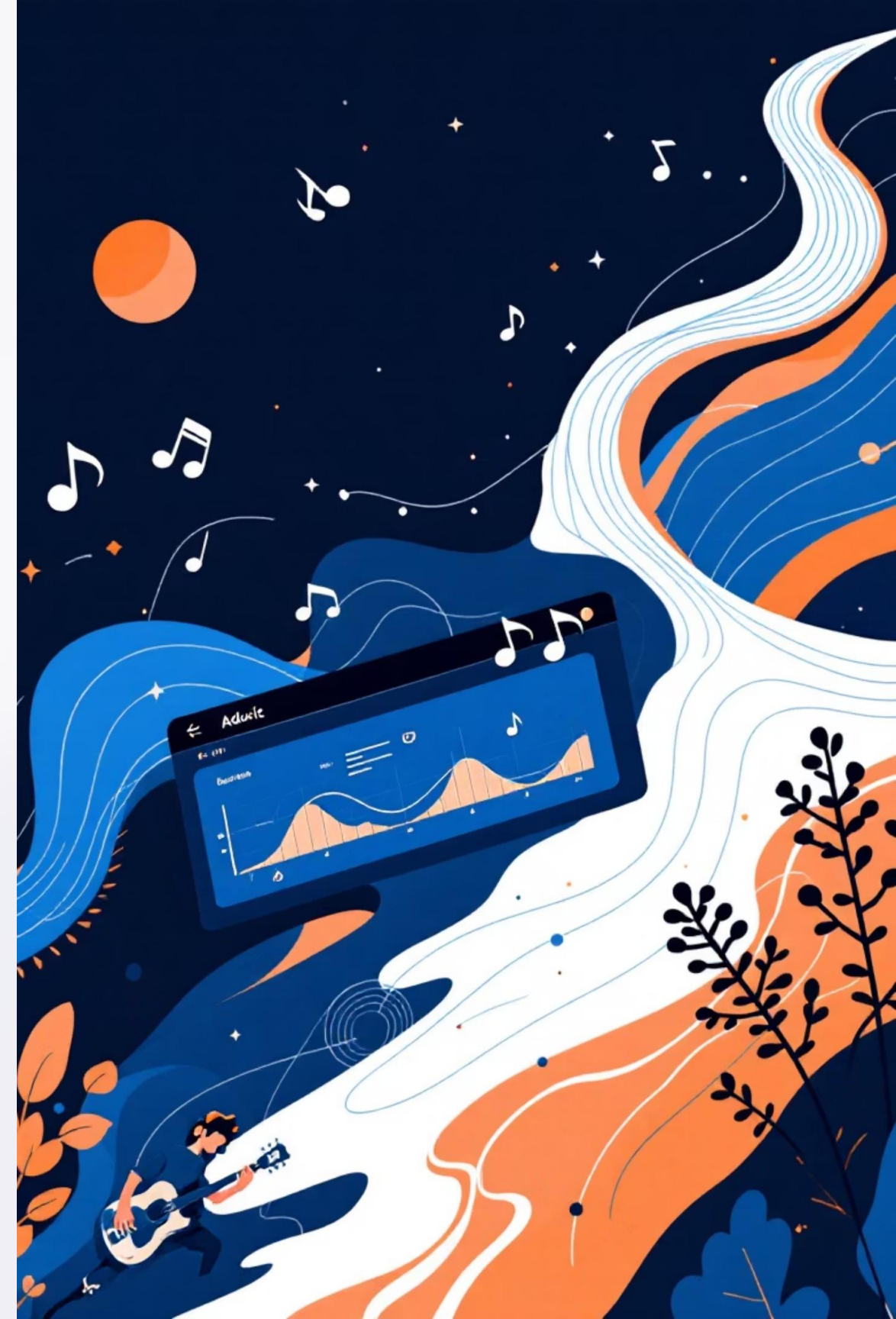
Dynamic User Profiling

Our system continuously builds a behavioral profile, learning from your completion rates and consistency to refine your path.

Evolving Guidance

Practice Arena emulates a human teacher, providing strategic, evolving guidance that deepens over time.

Unlike other platforms, Practice Arena synthesizes emotional context, time constraints, and behavioral data to craft a truly personalized and effective long-term practice journey.



Functional Requirements

Core Features

- User profile creation with instrument, level, and goals
- Mood selection before each session
- Time availability input
- Personalized practice plan generation
- Structured session segments (warm-up, technique, application)
- Task completion tracking
- Post-session feedback collection

Advanced Capabilities

- Progress tracking and visualization
- Practice history access
- External learning resource recommendations
- Daily music theory and ear-training game
- Dynamic user profiling with behavioral inference
- Long-term practice path (PATH) management
- LLM-assisted decision support using RAG



Non-Functional Requirements



Fast & Responsive

Practice plans generate in under two seconds, keeping you in the flow



Secure & Reliable

Your data is protected with access controls and reliable storage



Scalable

Designed to support a growing user base without performance issues



Modular & Maintainable

Easy to update recommendation logic and add new features



Transparent

Recommendation logic is explainable—you'll understand why tasks are suggested



Accessible Anywhere

Works in any standard web browser, no special software required

How It Works

01

Expert-Curated Practice Data

We manually create structured tables of exercises, techniques, and teaching rules based on real guitar instruction experience

02

Context Collection

You input your mood, available time, and goals—the system combines this with your practice history

03

Intelligent Algorithm

A rule-based recommendation engine selects and sequences tasks, balancing challenge with motivation

04

Behavioral Learning

The system tracks completion rates, consistency, and patterns to gradually refine your personalized practice path

05

LLM Decision Support

An AI assistant provides strategic guidance by reasoning over pedagogical rules and your progress trends

The result: practice sessions that feel like they were designed by a teacher who knows you personally—because in a way, they were.

System Architecture



Client Side (GUI)

The interactive user interface responsible for displaying practice plans, capturing user input (mood, time, goals), and communicating with the backend via the REST API.



REST API

Serves as the secure interface between the client application and the backend services, handling data requests, validation, authentication, and routing.



Backend & Database

Stores the core application logic, manages user profiles, stores practice history, and maintains the extensive library of expert-curated exercise data.



Recommendation & Behavioral Algorithm

A rule-based engine that processes user data, practice history, and pedagogical rules to generate personalized, balanced, and motivating practice task sequences.



LLM-Assisted Decision Support (RAG)

Utilizes Retrieval-Augmented Generation (RAG) to reason over pedagogical guidelines and user progress, providing strategic guidance and refining recommendations with contextual insights.

How the Algorithm Works

Practice Arena's algorithm uses a hybrid approach, combining rule-based logic with dynamic behavioral insights to create truly personalized practice sessions.



User & Context Input

Gathers static user profile, current mood, available time, and long-term practice path (PATH) for each session.



Rule Engine Logic

A deterministic engine selects and sequences exercises into structured segments based on expert-defined pedagogical rules.



Behavioral Data Capture

Collects implicit behavioral data during practice, such as task completion, time spent, and session consistency.



Adaptive Learning Loop

Analyzes behavioral signals to infer engagement and tolerance, making incremental, motivation-preserving adjustments.



LLM Advisory Module

An LLM-assisted RAG module provides pedagogical guidance for long-term decisions, ensuring explainable, rule-based outcomes.

This dynamic process ensures sessions are always relevant, challenging, and supportive, adapting to your evolving needs as a musician.

User Dashboard: History & Progress



Continuous Overview

A clear view of your practice history, progress trends, and unique learning behaviors.



Key Statistics

Track practice frequency, session completion rates, streaks, and total time spent practicing.



Progress Trends

Visualize improvement through difficulty trends and task completion patterns over time.



Session Review

Review past sessions, including assigned tasks, actual time spent, and post-session feedback.



Aggregated Insights

Intuitive insights on consistency, preferred categories, and fatigue indicators to support self-awareness.



Feedback Loop

Reinforces progress, highlights long-term development, and strengthens commitment to regular practice.

The dashboard acts as your personal practice mentor, providing a reflective space to understand and improve your musical journey.